HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79MA

STATISTICAL MODELS A (SOLUTION)

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2024-2025

Duration: 2 Hours

Attempt ALL questions.
Appropriate working must be shown to obtain full credit.

Question 1	17
Question 2	11
Question 3	15
Question 4	12
Question 5	5
Total	60

Resources Included within Exam Paper: Standard Formula sheet

Additional Resources to be Supplied to Candidates: New Cambridge Statistical Tables (Blue)

Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

1. (a) Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be an i.i.d. random sample from a uniform $\mathrm{U}(0, \theta)$ distribution, having probability density function given by

$$f(x; \theta) = \frac{1}{\theta}, \quad 0 < x < \theta,$$

with $\theta > 0$ being an unknown parameter.

[**Hint**: For the uniform distribution on (0, θ), the jth order statistic $\frac{X_{(j)}}{\theta} \sim \mathrm{Beta}(j, n + 1 - j)$]

i. Estimate θ using the method of moments (MME) and maximum likelihood estimation (MLE), denoting the estimators as $\tilde{\theta}$ and $\hat{\theta}$, respectively.

(3 marks)

Solution:
The mean of a uniform distribution $\mathrm{U}(0, \theta)$ is given by:

$$E(X) = \frac{\theta}{2}$$

To estimate θ using the MME, we set the sample mean equal to the expected value:

$$\bar{X} = \frac{\theta}{2}$$

Solve for θ, we get $\tilde{\theta} = 2\bar{X}$. **[+1 mark]**

To derive MLE, the likelihood function $L(\theta; \underline{X})$ is:

$$L(\theta; \underline{X}) = \prod_{i=1}^{n} f(X_i; \theta) = \prod_{i=1}^{n} \frac{1}{\theta} = \left(\frac{1}{\theta}\right)^n, \quad \text{for } 0 < X_i < \theta.$$

This can be rewritten as

$$L(\theta; \underline{X}) = \left(\frac{1}{\theta}\right)^n \cdot \mathrm{I}(\min x_i > 0) \cdot \mathrm{I}(\max x_i < \theta)$$

where I denotes the indicator function.

Clearly, all x_i's are positive, then there is no need for the differentiation, and a direct inspection of the likelihood reveals that the MLE is $\hat{\theta} = X_{(n)} = \max X_i$. **[+2 marks]**

ii. Find the mean and variance of each of the MME estimator $\tilde{\theta}$ and the MLE estimator $\hat{\theta}$. Are both estimators unbiased?

(5 marks)

Solution:
For the MME estimator, we have $\tilde{\theta} = 2\bar{X}$. The expected value of $\tilde{\theta}$ is calculated as follows:

$$\mathbb{E}(\tilde{\theta}) = \mathbb{E}(2\bar{X}) = 2\mathbb{E}(\bar{X}) = 2\left(\frac{\theta}{2}\right) = \theta.$$

$$\therefore \tilde{\theta} \text{ is an unbiased estimator of } \theta. \quad \textbf{[+1 mark]}$$

Next, we compute the variance of $\tilde{\theta}$:

$$\mathrm{Var}(\tilde{\theta}) = \mathrm{Var}(2\bar{X}) = 4\cdot\mathrm{Var}(\bar{X}) = \frac{4}{n}\mathrm{Var}(X) = \frac{4}{n}\left(\frac{\theta^2}{12}\right) = \frac{\theta^2}{3n}. \quad (\because \mathrm{Var}(X) = \frac{\theta^2}{12}) \quad \textbf{[+1 mark]}$$

For the MLE estimator, $\hat{\theta} = X_{(n)}$ (the maximum order statistic), we know that $\frac{X_{(n)}}{\theta} \sim \mathrm{Beta}(n, 1)$.
The expected value of $\frac{X_{(n)}}{\theta}$ is:

$$\mathbb{E}\left(\frac{X_{(n)}}{\theta}\right) = \frac{n}{n+1}.$$

Thus,

$$\mathbb{E}(\hat{\theta}) = \mathbb{E}(X_{(n)}) = \frac{n}{n+1}\theta \neq \theta.$$

$$\therefore \hat{\theta} \text{ is a biased estimator of } \theta. \quad \textbf{[+1 mark]}$$

Now, for the variance of $\hat{\theta}$, we use the fact that:

$$\mathrm{Var}\left(\frac{X_{(n)}}{\theta}\right) = \frac{n}{(n+1)^2(n+2)}.$$

Thus,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}(X_{(n)}) = \frac{n}{(n+1)^2(n+2)}\theta^2. \quad \textbf{[+2 marks]}$$

iii. Derive the mean square errors (MSE) of each of the MME estimator $\tilde{\theta}$ and the MLE estimator $\hat{\theta}$. Which estimator should be preferred? Explain.

(3 marks)

Solution:
For the method of moments estimator (MME):

$$\text{MSE}(\tilde{\theta}) = \text{Var}(\tilde{\theta}) = \frac{\theta^2}{3n} \quad \textbf{[+1 mark]}$$

since $\tilde{\theta}$ is unbiased and has zero bias.

For the maximum likelihood estimator (MLE):

$$B(\hat{\theta}) = \left(\frac{n}{n+1}\right)\theta - \theta = -\frac{\theta}{n+1}.$$

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + B(\hat{\theta})^2 = \frac{n\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{\theta^2}{(n+1)^2}\left(\frac{n}{n+2}+1\right) = \frac{2\theta^2}{(n+1)(n+2)}.$$

Preferred Estimator Comments: (Give **[+1 mark]** for sensible comments)

- The MME, $\tilde{\theta}$, is unbiased; however, its MSE is generally larger than that of the MLE, $\hat{\theta}$, especially for large n. While the MSE of $\tilde{\theta}$ is consistent and decreases as n increases, it does so more slowly than the MSE of $\hat{\theta}$.
- The MLE, $\hat{\theta}$, is biased but gives a smaller MSE compared to the MME for moderate to large sample sizes. As n large, the bias of $\hat{\theta}$ approaches to zero, indicating that it is an asymptotically unbiased estimator.
- For small sample sizes, the MME may be preferred due to its unbiased nature, as the MSEs of both estimators are likely to be close or equal.
- For moderate to large sample sizes, the MLE is generally preferred due to its smaller MSE, which converges to zero faster than that of the MME. Additionally, it is asymptotically unbiased and consistent for estimating θ.

iv. Determine an expression for a confidence interval for θ using the MME $\tilde{\theta} = 2\bar{X}$, assuming a Normal approximation to the distribution of $\tilde{\theta}$.

(3 marks)

Solution:
Since the MME is $\tilde{\theta} = 2\bar{X}$, we know that:

$$E[\tilde{\theta}] = 2E[\bar{X}] = 2 \times \frac{\theta}{2} = \theta,$$

and

$$\text{Var}(\tilde{\theta}) = 4 \times \text{Var}(\bar{X}) = 4 \times \frac{\theta^2}{12n} = \frac{\theta^2}{3n}.$$

Thus, $\tilde{\theta}$ is approximately normally distributed for large n:

$$\tilde{\theta} \sim \mathcal{N}\left(\theta, \frac{\theta^2}{3n}\right). \qquad \textbf{[+1 mark]}$$

Therefore,

$$P\left(-z_{\alpha/2} < \frac{\tilde{\theta} - \theta}{\theta/\sqrt{3n}} < z_{\alpha/2}\right) = 1 - \alpha,$$

Substituting the MME estimator $\theta = \tilde{\theta} = 2\bar{X}$ into the denominator and rearranging the expression, the $100(1-\alpha)\%$ confidence interval for θ using the MME is:

$$\left(2\bar{X} - z_{\alpha/2} \cdot \frac{2\bar{X}}{\sqrt{3n}},\ 2\bar{X} + z_{\alpha/2} \cdot \frac{2\bar{X}}{\sqrt{3n}}\right)$$

([**+2 marks**] for a reasonable CI for θ using the MME)

[Alternatively, after substituting the MME estimator $\tilde{\theta} = 2\bar{X}$, the CI can be expressed as:

$$P\left(\frac{-z_{\alpha/2}}{\sqrt{3n}} < \frac{2\bar{X}}{\theta} - 1 < \frac{z_{\alpha/2}}{\sqrt{3n}}\right) = 1 - \alpha.$$

Rearranging gives the $100(1-\alpha)\%$ CI for θ using the MME is:

$$\left(\frac{2\bar{X}}{1 + \frac{z_{\alpha/2}}{\sqrt{3n}}}, \frac{2\bar{X}}{1 - \frac{z_{\alpha/2}}{\sqrt{3n}}}\right)]$$

(b) Suppose that $\underline{X} = (X_1, X_2, \ldots, X_n)$ is an i.i.d. random sample from a Beta distribution with parameters α and β, where α is known and β is unknown ($\beta > 0$).
The probability density function of the Beta distribution is given by:

$$f(x; \alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1,$$

where $\Gamma(\cdot)$ represents the Gamma function.
Show that

$$T = \frac{\left(\sum_{i=1}^{n} \log(1-X_i)\right)^3}{n}$$

is a sufficient statistic for the parameter β.

(3 marks)

Solution:
Given α is known, the likelihood is given by

$$\begin{aligned} L(\beta) = f_{\underline{X}}(\underline{x}; \beta) &= \prod_{i=1}^{n} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x_i^{\alpha-1}(1-x_i)^{\beta-1} \\ &= \left(\frac{1}{\Gamma(\alpha)^n}\right)\left(\prod_{i=1}^{n} x_i^{\alpha-1}\right) \times \left(\frac{\Gamma(\alpha+\beta)}{\Gamma(\beta)} \prod_{i=1}^{n}(1-x_i)^{\beta-1}\right) \\ &= \left(\frac{1}{\Gamma(\alpha)^n}\right)\left(\prod_{i=1}^{n} x_i^{\alpha-1}\right) \times \left(\left(\frac{\Gamma(\alpha+\beta)}{\Gamma(\beta)}\right)^n e^{\sum_{i=1}^{n} \log(1-x_i)^{\beta-1}}\right) \\ &= h_1(\underline{x}) \times h_2\left(\beta, \sum_{i=1}^{n} \log(1-x_i)\right). \end{aligned}$$

Hence, by the factorisation theorem, $\sum \log(1-X_i)$ is sufficient for β. **[+2 marks]**
Since, $T = \frac{\left(\sum_{i=1}^{n} \log(1-X_i)\right)^3}{n}$ is a one-to-one function of the sufficient statistic $\sum \log(1-X_i)$, it follows that T is also sufficient for β. **[+1 mark]**

[Total 17 marks]

2. Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be an i.i.d. random sample from a distribution having probability density function given by

$$f(x;\theta) = \frac{\theta^{1/3}}{\Gamma(4/3)} \exp(-\theta x^3), \quad 0 < x < \infty,$$

where $\theta > 0$ is an unknown parameter and $\Gamma(z)$, as usual, denotes the Gamma function given by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt.$$

(a) Show that the score function $U(\theta)$ is given by

$$U(\theta) = \frac{n}{3\theta} - \sum_{i=1}^{n} X_i^3.$$

(3 marks)

Solution:
First we find the likelihood $L(\theta; \underline{X})$:

$$L(\theta; \underline{X}) = \prod_{i=1}^{n} f(X_i;\theta) = \prod_{i=1}^{n} \frac{\theta^{1/3}}{\Gamma(4/3)} \exp(-\theta X_i^3) = \frac{\theta^{n/3}}{(\Gamma(4/3))^n} \exp\left(-\theta \sum_{i=1}^{n} X_i^3\right).$$

[+1 mark]
Taking logs gives the log-likelihood $l(\theta; \underline{X})$:

$$l(\theta; \underline{X}) = \frac{n}{3}\log(\theta) - n\log(\Gamma(4/3)) - \theta \sum_{i=1}^{n} X_i^3.$$

[+1 mark]
Taking the derivative with respect to θ gives the score function

$$U(\theta) = \frac{\partial l}{\partial \theta} = \frac{n}{3\theta} - \sum_{i=1}^{n} X_i^3.$$

[+1 mark]

(b) Find the maximum likelihood estimator $\hat{\theta}$ of θ.

(1 mark)

Solution:
Setting $U(\hat{\theta}) = 0$ gives

$$\hat{\theta} = \frac{n}{3\sum_{i=1}^{n} X_i^3}.$$

[+1 mark]

(c) Without doing any calculations, is $\hat{\theta}$ unbiased? Give a reason for your answer.

(2 marks)

Solution:
$\hat{\theta}$ must be biased since it is a smooth estimator and we know from the Corollary proven in class that if an MLE $\hat{\theta}$ is unbiased we can write the score function as

$$U(\theta) = I(\theta)(\hat{\theta} - \theta)$$

which we clearly cannot do here.
[+2 marks]

(d) Calculate the Fisher information $I(\theta)$ and hence write down the asymptotic distribution of $\hat{\theta}$.

(2 marks)

Solution:

$$I(\theta) = -\mathbf{E}\left(\frac{\partial U}{\partial \theta}\right) = -\mathbf{E}\left(-\frac{n}{3\theta^2}\right) = \frac{n}{3\theta^2}.$$

[+1 mark]
By the maximum likelihood theorem the asymptotic distribution of the MLE $\hat{\theta}$ is given by

$$\frac{\hat{\theta} - \theta}{\sqrt{1/I(\theta)}} \approx N(0, 1).$$

Hence we have

$$\frac{\hat{\theta} - \theta}{\sqrt{3\theta^2/n}} \approx N(0, 1).$$

[+1 mark]

(e) Suppose you are given a sample $\underline{x} = (x_1, \ldots, x_{15})$ of size 15 such that

$$\sum_{i=1}^{15} x_i^3 = 5.$$

Calculate $\hat{\theta}$ for this sample and hence show that with this estimated value of θ we have

$$\mathbf{E}(X^3) = \frac{\sum_{i=1}^{n} x_i^3}{n}.$$

Hint: You may find it helpful to consider the definition of $\Gamma(4/3)$

(3 marks)

Solution:
Here

$$\hat{\theta} = \frac{15}{3 \times 5} = 1.$$

[+1 mark]

$$\begin{aligned}
\mathbf{E}(X^3) &= \int_{x=0}^{\infty} x^3 \frac{\theta^{1/3}}{\Gamma(4/3)} \exp(-\theta x^3) \mathrm{dx} \\
&\approx \int_{x=0}^{\infty} \frac{x^3}{\Gamma(4/3)} \exp(-x^3) \mathrm{dx} \quad (\text{using } \theta \approx 1) \\
&= \int_{t=0}^{\infty} \frac{t^{1/3}}{3\Gamma(4/3)} \exp(-t) \mathrm{dt} \quad (\text{using } t = x^3) \\
&= \frac{1}{3\Gamma(4/3)} \int_{t=0}^{\infty} t^{4/3-1} e^{-t} \mathrm{dt} \quad (\text{using } t = x^3) \\
&= \frac{1}{3\Gamma(4/3)} \Gamma(4/3) = \frac{1}{3}.
\end{aligned}$$

[+2 marks]

[Total 11 marks]

3. Let $X \sim N(0, \sigma^2)$, and consider testing $H_0 : \sigma = \sigma_0$ versus $H_1 : \sigma = \sigma_1$, where $\sigma_1 > \sigma_0$. The values σ_0 and σ_1 are fixed.

   (a) Find an expression for the likelihood ratio as a function of x. Hence, identify the values of x that favour H_0 and specify a rejection region for a level α test.

(4 marks)

Solution: We are given that $X \sim N(0, \sigma^2)$. The likelihood function for x is:

$$L(\sigma^2; x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

We are testing the hypotheses:

- $H_0 : \sigma = \sigma_0$
- $H_1 : \sigma = \sigma_1$, where $\sigma_1 > \sigma_0$

The likelihood ratio $LR(x)$ is given by:

$$LR(x) = \frac{L(\sigma_0^2; x)}{L(\sigma_1^2; x)} = \frac{\frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{x^2}{2\sigma_0^2}\right)}{\frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{x^2}{2\sigma_1^2}\right)}.$$

Simplifying the likelihood ratio:

$$LR(x) = \frac{\sigma_1}{\sigma_0} \exp\left(-\frac{x^2}{2}\left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}\right)\right).$$

[+2 marks]

To favour H_0, we look for values of x that make the likelihood ratio $LR(x)$ large. Since the exponential term depends on x^2, smaller values of x^2 will result in a larger $LR(x)$. We conclude that small values of x^2 (or small values of $|x|$) favour H_0.

[+1 mark]

According to the Neyman-Pearson Lemma, the rejection region R is defined as:

$$R = \{x : \ LR(x) \leq k\}$$

which may be equivalently defined as

$$R = \left\{x : \ x^2 \geq k_1\right\}$$

with k_1 set such that the Type I error probability $\mathbf{P}(X \in R \mid \sigma = \sigma_0) = \alpha$.

Since $\frac{X}{\sigma} \sim N(0, 1)$, we have $\left(\frac{X}{\sigma}\right)^2 \sim \chi_1^2$. Hence, the threshold of the rejection region R is

$$k_1 = \sigma_0^2 \chi_1^2(\alpha)$$

where $\chi_1^2(\alpha)$ is the $(1 - \alpha)$-quantile of the chi-squared distribution with 1 degree of freedom.

[+1 mark]

(b) For an i.i.d. random sample of $X_1, X_2, \ldots, X_n$ distributed as $N(0, \sigma^2)$, derive the likelihood ratio and the rejection region for the same hypotheses as in part (a).

(4 marks)

Solution: The likelihood ratio for a sample $X_1, X_2, \ldots, X_n$ is given by:

$$LR(\underline{x}) = \frac{L(\sigma_0^2; \underline{x})}{L(\sigma_1^2; \underline{x})} = \left(\frac{\sigma_1}{\sigma_0}\right)^n \exp\left(-\frac{1}{2}\left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}\right)\sum_{i=1}^n x_i^2\right).$$

[+2 marks]
According to the Neyman-Pearson Lemma, the rejection region of the test is:

$$\left\{\underline{x} : \sum_{i=1}^n x_i^2 \geq k_2\right\},$$

with k_2 set such that the Type I error probability $\mathbf{P}(\underline{X} \in R \mid \sigma = \sigma_0) = \alpha$.

Since $\left(\frac{X}{\sigma}\right)^2 \sim \chi_1^2$, we have $\sum \left(\frac{X}{\sigma}\right)^2 \sim \chi_n^2$. Hence, the threshold of the rejection region is

$$k_2 = \sigma_0^2 \chi_n^2(\alpha)$$

where $\chi_n^2(\alpha)$ is the $(1-\alpha)$-quantile of the chi-squared distribution with n degrees of freedom.
[+2 marks]

(c) Is the likelihood ratio test derived in part (b) uniformly most powerful for testing $H_0 : \sigma = \sigma_0$ against $H_1 : \sigma > \sigma_0$?

(4 marks)

Solution:
Method 1:
We note that the likelihood ratio

$$LR(\underline{x}) = \left(\frac{\sigma_1}{\sigma_0}\right)^n \exp\left(-\frac{1}{2}\left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}\right)\sum_{i=1}^n x_i^2\right)$$

is a monotonically decreasing function in the sufficient statistic $\sum x_i^2$ for all $\sigma_1 > \sigma_0$. Hence, the distribution $N(0, \sigma^2)$ has the Monotone Likelihood Ratio Property (MLRP).

Method 2:
We say that a random variable has a distribution from the exponential family if we can write its probability density function in the form:

$$f(x; \theta) = c(\theta)h(x)\exp\{q(\theta)t(x)\}$$

for some functions c, h, q, and t.

Since the probability density function of $X \sim N(0, \sigma^2)$ can be written as

$$\frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2} \cdot x^2\right),$$

X belongs to the exponential family of distributions. In this case, the test statistic $\sum x_i^2$ is sufficient for σ^2.
Since $q(\sigma) = -1/(2\sigma^2)$ is monotonic decreasing in σ, it follows that the MLRP is satisfied through $\sum x_i^2$.
[+2 marks]

Conclusion:
From either of the above arguments, the likelihood ratio test derived in part (b) is uniformly most powerful (UMP) for testing $H_0 : \sigma = \sigma_0$ against $H_1 : \sigma > \sigma_0$, as guaranteed by the Neyman-Pearson Lemma. The test maximises power for all alternatives $\sigma > \sigma_0$.
[+2 marks]

(d) Perform numerical calculations for the following scenario: Consider a sample of size $n = 10$, with $\sigma_0 = 1$, $\sigma_1 = 1.5$, and an observed value $\sum_{i=1}^{10} x_i^2 = 14$. At a significance level of $\alpha = 0.05$, does the test reject H_0?

(3 marks)

Solution: First, calculate the critical value from the chi-squared distribution with $n = 10$ degrees of freedom at $\alpha = 0.05$:

$$\chi^2_{10}(0.05) = 18.31.$$

[+1 mark]
The rejection region is:

$$\sum_{i=1}^{10} x_i^2 > 1^2 \times 18.31 = 18.31.$$

Since the observed value $\sum_{i=1}^{10} x_i^2 = 14$ is less than 18.31, we fail to reject H_0 at the $\alpha = 0.05$ level. **[+2 marks]**

[Alternatively, students may calculate the p-value as

$$\mathbf{P}\left(\sum_{i=1}^{10} X_i^2 \geq 14\right) = 0.173.$$

Since the p-value $> 0.05 = \alpha$, there is not enough evidence to reject H_0.]

[Total 15 marks]

4. Suppose that you work as an actuary at an insurance company. As an actuary, you are interested in understanding the arrival times of insurance claims at the company. Denote by X the arrival time of an insurance claim. You are told that the times at which the insurance claims arrive at the company follow a continuous distribution with the following density function:

$$f_X(x) = 3\theta x^2 e^{-\theta x^3}, \text{ for } x > 0,$$

where $\theta > 0$ is a rate parameter.

Suppose that you have observed times $x_1, x_2, \cdots, x_n$ for n insurance claims throughout the year. To perform Bayesian analyses, you decide to use the prior distribution $\theta \sim \text{Gamma}(\alpha, \beta)$. Answer the following questions:

(a) Show that the posterior distribution of $(\theta|\underline{\text{x}})$ also follows the gamma distribution. State clearly the parameters of the distribution of $(\theta|\underline{\text{x}})$. What is the Bayes estimate for θ under the minimisation of the mean square error?

(4 marks)

Solution:

From Bayes formula, the posterior density for $(\theta|\underline{\text{x}})$ is proportional to

$$\begin{aligned}\pi(\theta|\underline{\text{x}}) &\propto \prod_{i=1}^{n} \theta x_i^2 e^{-\theta x_i^3} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} \\ &\propto \theta^{n+\alpha-1} e^{-\left(\beta + \sum x_i^3\right)\theta}\end{aligned}$$

which is proportional to the pdf of Gamma$(n+\alpha, \beta + \sum x_i^3)$. **[+2 marks]**

We know that $\mathbb{E}(\theta|\underline{\text{x}})$ is the mean of the posterior distribution, i.e., Bayes estimate with respect to the mean square error loss.

$$\mathbb{E}(\theta|\underline{\text{x}}) = \frac{n+\alpha}{\beta + \sum x_i^3}. \quad \textbf{[+2 marks]}$$

(b) Express the Bayes estimate found in part (a) in the form of $ZT + (1-Z)\mu$, where $0 \le Z \le 1$ is a weighting factor, T is a statistic of $\underline{\text{x}}$ (or a transformation of the statistic), and μ is a component from our prior information. State clearly the values of Z, T, and μ. Also, describe the statistic T in words.

(5 marks)

Solution:

Note that we can express the Bayes estimate as

$$\begin{aligned}\mathbb{E}(\theta|\underline{\text{x}}) &= \frac{n+\alpha}{\beta + \sum x_i^3} \\ &= \frac{n}{\beta + \sum x_i^3} + \frac{\alpha}{\beta + \sum x_i^3} \\ &= \frac{\sum x_i^3}{\beta + \sum x_i^3} \cdot \frac{n}{\sum x_i^3} + \frac{\beta}{\beta + \sum x_i^3} \cdot \frac{\alpha}{\beta} \\ &= \frac{\sum x_i^3}{\beta + \sum x_i^3} \cdot \frac{1}{\frac{\sum x_i^3}{n}} + \left(1 - \frac{\sum x_i^3}{\beta + \sum x_i^3}\right) \cdot \frac{\alpha}{\beta} \quad \textbf{[+2 marks]}\end{aligned}$$

The values of Z, T, and μ are equal to

$$Z = \frac{\sum x_i^3}{\beta + \sum x_i^3},$$

$$T = \frac{n}{\sum x_i^3} \left(= \frac{1}{\frac{\sum x_i^3}{n}} \right),$$

and

$$\mu = \frac{\alpha}{\beta}$$

respectively. **[+2 marks]**

The statistic T is the reciprocal (or inverse) of the 3rd sample moment of $(X_1, X_2, \cdots, X_n)$ or any sensible comment. **[+1 mark]**

(c) As we collect more samples, i.e., as n grows larger, explain what would happen to the values of

i. the weighting factor Z,
ii. the component μ,
iii. the Bayes estimate for θ, $\mathbb{E}(\theta|\underline{x})$,

Justify your answers.

(3 marks)

Solution:

i. Since $x_i > 0 \ \forall i$, we have $\sum_{i=1}^n x_i^3 \to \infty$ as $n \to \infty$. Hence, the credibility factor Z approaches 1. **[+1 mark]**

ii. Since the component $\mu = \alpha/\beta$ is independent of n, the value of μ remains unchanged. **[+1 mark]**

iii. Note that

$$\mathbb{E}(\theta|\underline{x}) = Z \times \left(\frac{\sum x_i^3}{n} \right)^{-1} + (1 - Z) \times \frac{\alpha}{\beta}.$$

As $n \to \infty$, we have $Z \to 1$. Hence,

$$\mathbb{E}(\theta|\underline{x}) \to \left(\frac{\sum x_i^3}{n} \right)^{-1} \quad \textbf{[+1 mark]}$$

Alternatively, it is also acceptable to conclude that

$$\mathbb{E}(\theta|\underline{x}) \to \frac{1}{\mathbb{E}(X^3)},$$

by the law of large numbers, where $\mathbb{E}(X^3)$ is the 3rd population moment of X.

[Total 12 marks]

5. The table below shows the aggregate claims in five successive years from 4 insurance policies. The monetary amounts have been adjusted to account for the effects of inflation, and they are given in units of £1000.

Policy number (i)	Year (j) 1	2	3	4	5	$\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$	$S_i^2 = \frac{1}{n-1}\sum_{j=1}^{n}(X_{ij}-\bar{X}_i)^2$
1	70	85	73	91	88	81.4	87.3
2	95	90	82	100	91	91.6	44.3
3	61	42	50	67	37	51.4	158.3
4	54	51	53	60	57	55	12.5

Using the Empirical Bayes Credibility Theory Model 1 (Bühlmann model), calculate the credibility premium for the coming years for policy number 4.

[Total 5 marks]

Solution:
Let N be the number of policies and n be the number of years observed.

Since $m(\theta_i)$ is estimated by $\bar{X}_i$, we will estimate $\mathbb{E}[m(\theta_i)]$ by

$$\frac{1}{N}\sum_{i=1}^{N}\bar{X}_i = \frac{81.4+91.6+51.4+55}{4} = 69.85 \quad \textbf{[+1 mark]}$$

Since $s^2(\theta_i)$ is estimated by S_i^2, we will estimate $\mathbb{E}[s^2(\theta_i)]$ by

$$\frac{1}{N}\sum_{i=1}^{N}S_i^2 = \frac{87.3+44.3+158.3+12.5}{4} = 75.6 \quad \textbf{[+1 mark]}$$

The estimate for $\mathbb{V}\text{ar}[m(\theta_i)]$ is

$$\begin{aligned}
&\frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{X}_i-\bar{X}\right)^2 - \frac{1}{nN}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij}-\bar{X}_i\right)^2\right] \\
&= \frac{(81.4-69.85)^2+(91.6-69.85)^2+(51.4-69.85)^2+(55-69.85)^2}{3} - \frac{75.6}{5} \\
&= 374.01 \qquad \textbf{[+1 mark]}
\end{aligned}$$

The credibility factor Z has the following formula

$$Z = \frac{n}{n+\frac{\mathbb{E}[s^2(\theta_i)]}{\mathbb{V}\text{ar}[m(\theta_i)]}}$$

which is estimated as

$$Z = \frac{5}{5+\frac{75.6}{374.01}} = 0.961144 \qquad \textbf{[+1 mark]}$$

Hence, the credibility premium for the coming year for policy number 4 is

$$0.961144 \times 55 + (1 - 0.961144) \times 69.85 = 55.58 \text{ thousands} \quad \textbf{[+1 mark]}$$

Do not penalise the student for giving the final answer as 55.58.

END OF PAPER (TOTAL 60 marks)